

Final Therapy Plan: Digital Facial Expression Training for Parkinson's Patients (MG 20.01.2026)

Executive Summary

This evidence-based therapy plan defines a structured, progressive facial expression training program for Parkinson's patients with hypomimia. The program combines isolated and alternating exercises over an 8-week intensive cycle with a subsequent maintenance phase. The structure is based on motor learning principles, mastery-learning approaches, and realistic adherence research.

Core Features: - **9 core exercises** targeting different facial muscle groups - **Progressive hold duration:** 15 sec. → 20 sec. → 25 sec. - **Progressive complexity:** Isolated → Pairs → Sequences - **Realistic frequency:** 3-4×/week (Weeks 1-4), 3×/week (Weeks 5-8), 2-3×/week (Maintenance) - **Session duration:** 18-27 minutes - **Avatar-guided implementation** with 111 animation sequences

Part 1: Program Overview

Overall Structure

Phase	Weeks	Frequency	Hold Duration	Exercise Type	Session	Weekly Load
1: Foundations	1-2	3-4×/week	15 sec.	Isolated Exercises	18-20 min.	54-80 min.
2: Consolidation	3	3-4×/week	20 sec.	Antagonistic Pairs	22-24 min.	66-96 min.
3: Intensification	4	3-4×/week	25 sec.	Complex Sequences	25-27 min.	75-108 min.
4: Mastery	5-8	3×/week	25 sec.	Mix (60% Seq. + 40% Iso.)	25-27 min.	75-81 min.
Maintenance	9+	2-3×/week	25 sec.	Flexible Mix	15-20 min.	30-60 min.

The 9 Core Exercises

No.	Exercise	Muscular Activation	Target Muscles	FACS AU
1	Angry	Eyebrows together, mouth clenched	M. corrugator supercilii, M. orbicularis oris	AU4, AU7, AU23, AU24
2	Surprised	Eyes wide open, mouth wide open	M. frontalis, M. levator palpebrae	AU1, AU2, AU5, AU26
3	Disgust	Eyebrows drawn together, upper lip raised	M. nasalis, M. levator labii superioris	AU9, AU15
4	Fear	Eyes squeezed, mouth clenched	M. frontalis, M. corrugator, M. platysma	AU1, AU2, AU4, AU5, AU7, AU20
5	Smile	Mouth drawn wide	M. zygomaticus major, M. orbicularis oculi	AU12, AU6
6	Sadness	Mouth corners downward	M. depressor anguli oris	AU1, AU4, AU15
7	Pout	Mouth pursed	M. orbicularis oris	AU18, AU24
8	Puff cheeks	Cheeks inflated	M. buccinator (relaxed)	-
9	Smile + Cheeks	Combined exercise	M. zygomaticus + M. buccinator	AU12 + Cheeks

Part 2: Detailed Phase Descriptions

Phase 1: Foundations (Week 1-2)

Goal: Establish correct movement patterns, learn muscle activation

Parameters: - **Frequency:** 3-4× per week - **Hold duration:** 15 seconds - **Repetitions:** 3× per exercise - **Rest between repetitions:** 10 seconds - **Session duration:** 18-20 minutes - **Exercise type:** Isolated exercises (Blocked Practice)

Daily Structure:

WARM-UP (3 Minutes)	
• Facial massage: Forehead, cheeks, jaw (90 sec.) • Gentle grimacing (60 sec.) • Conscious breathing (30 sec.)	

AVATAR INTRODUCTION (1 Minute)	
"Today we practice 9 facial expressions. Each expression will be held for 15 seconds. Focus on correct muscle activation."	
MAIN EXERCISE BLOCK (14 Minutes)	
Exercise 1: Angry - Avatar demonstrates (5 sec.) 1st repetition: 15 sec. → 10 sec. rest 2nd repetition: 15 sec. → 10 sec. rest 3rd repetition: 15 sec. → 10 sec. rest - Brief feedback (5 sec.) Total: ~1.5 minutes [Same structure for Exercises 2-9] Exercise 2: Surprised (1.5 min.) Exercise 3: Disgust (1.5 min.) Exercise 4: Fear (1.5 min.) Exercise 5: Smile (1.5 min.) Exercise 6: Sadness (1.5 min.) Exercise 7: Pout (1.5 min.) Exercise 8: Puff cheeks (1.5 min.) Exercise 9: Smile + Cheeks (1.5 min.)	
COOL DOWN (2 Minutes)	
- Gently tap face (60 sec.) - Relaxation exercise (30 sec.) - Conscious smile as closing ritual (30 sec.)	
SESSION COMPLETION (1 Minute)	
Avatar shows results: "You performed 8 of 9 exercises correctly (89%)" "Particularly good: Surprised, Smile, Pout" "Next time we'll focus on: Disgust"	

TOTAL DURATION: 18-20 Minutes

Mastery Criterion Phase 1: - ✓ ≥80% of exercises correct (at least 7 of 9) - ✓ Average hold duration ≥12 seconds (80% of 15 sec.) - ✓ Patient reports familiarity with all expressions → **PROGRESSION to Phase 2**

If Criterion Not Met: - Repeat Week 2 - Optional: Focus on 6 best exercises - Optional: Reduce hold duration to 12 sec.

Phase 2: Consolidation with Antagonistic Pairs (Week 3)

Goal: Train reciprocal inhibition, increase contextual interference

Parameters: - **Frequency:** 3-4× per week - **Hold duration:** 20 seconds - **Repetitions:** 4× per pair (alternating) - **Rest between emotions:** 5 seconds - **Rest after pair:** 10 seconds - **Session duration:** 22-24 minutes - **Exercise type:** Antagonistic pairs (Semi-Random Practice)

Antagonistic Pairings:

Pair 1: Surprised ↔ Angry - Antagonism: M. frontalis (raise forehead) ↔ M. corrugator supercilii (frown forehead)

Pair 2: Smile ↔ Sadness - Antagonism: M. zygomaticus major (mouth corners up) ↔ M. depressor anguli oris (mouth corners down)

Pair 3: Disgust ↔ Neutral - Focus: Conscious relaxation after muscle activation

Pair 4: Fear ↔ Surprised - Different intensities of eye opening

Daily Structure:

WARM-UP (3 Minutes) - as in Phase 1	
-------------------------------------	--

AVATAR INTRODUCTION (1 Minute)	
--------------------------------	--

"This week we practice emotions in pairs." "You will alternate between two expressions." "Hold duration: 20 seconds. Focus on smooth transitions between emotions."	
---	--

MAIN EXERCISE BLOCK (18-20 Minutes)	
-------------------------------------	--

BLOCK 1: Pair 1 - Surprised ↔ Angry (5 min.) - Avatar demonstrates sequence (10 sec.) - Patient:	
--	--

Neutral → SURPRISED (20s) → Rest (5s) → ANGRY (20s) → Rest (10s) Repeat: SURPRISED → ANGRY → SURPRISED → ANGRY (4× each expression alternating) └ Return to neutral (10 sec.)	
REST (1 Minute)	
BLOCK 2: Pair 2 - Smile ↔ Sadness (5 min.) └ Same structure as Pair 1	
REST (1 Minute)	
BLOCK 3: Pair 3 - Disgust ↔ Neutral (4 min.) └ Repeat 4× (focus on active relaxation)	
BLOCK 4: Isolated Exercises (5-6 min.) └ Pout: 20 sec., 3× (1.5 min.) └ Puff cheeks: 20 sec., 3× (1.5 min.) └ Smile + Cheeks: 20 sec., 3× (1.5 min.) └ Fear: 20 sec., 3× (1.5 min.)	

COOL DOWN (2 Minutes) - as in Phase 1	
---------------------------------------	--

SESSION COMPLETION (1 Minute)	
"You mastered 4 emotion pairs today!" "Success rate: 85%" "Transition quality: Good" "You're ready for complex sequences!"	

TOTAL DURATION: 22-24 Minutes

Mastery Criterion Phase 2: - ✓ ≥85% of alternating pairs correct - ✓ Average hold duration ≥16 seconds (80% of 20 sec.) - ✓ Smooth transitions (4-6 sec. transition time)
→ **PROGRESSION to Phase 3**

If Criterion Not Met: - Repeat Week 3 - Optional: Only 2 pairs instead of 4 - Optional: Increase transition time to 7 seconds

Phase 3: Intensification with Complex Sequences (Week 4)

Goal: Maximum contextual interference, train realistic emotion transitions

Parameters: - **Frequency:** 3-4× per week - **Hold duration:** 25 seconds (target duration reached!) - **Repetitions:** 2× per sequence - **Transition time:** 5 seconds between emotions - **Rest after sequence:** 20 seconds - **Session duration:** 25-27 minutes - **Exercise type:** Complex multi-emotion sequences (Random Practice)

Predefined Sequences:

Sequence A: Upper Face - Neutral → Surprised (25s) → Angry (25s) → Surprised (25s) → Neutral - Duration: ~2.5 min. per pass

Sequence B: Lower Face - Neutral → Smile (25s) → Sadness (25s) → Pout (25s) → Smile (25s) → Neutral - Duration: ~3 min. per pass

Sequence C: Whole Face - Neutral → Fear (25s) → Disgust (25s) → Surprised (25s) → Neutral - Duration: ~2.5 min. per pass

Sequence D: Special Exercises - Neutral → Puff cheeks (25s) → Smile + Cheeks (25s) → Neutral - Duration: ~1.5 min. per pass

Daily Structure:

WARM-UP (3 Minutes)	
AVATAR INTRODUCTION (1 Minute)	
"Welcome to Week 4!" "Today we practice complex emotion sequences." "Hold duration: 25 seconds - our target duration!"	
MAIN EXERCISE BLOCK (20-22 Minutes)	
BLOCK 1: Sequence A - Upper Face (6 min.) - Avatar demonstrates slowly (30 sec.) 1st pass: - Neutral → SURPRISED (25s) → [5s transition] → ANGRY (25s) → [5s] → SURPRISED (25s) → Neutral - Rest (20 sec. - cognitive recovery) 2nd pass: Repeat	
REST (2 Minutes)	

BLOCK 2: Sequence B - Lower Face (7 min.)	
└ 1st pass:	
Neutral → SMILE (25s) → SADNESS (25s)	
→ POUT (25s) → SMILE (25s) → Neutral	
└ Rest (20 sec.)	
└ 2nd pass	
REST (2 Minutes)	
BLOCK 3: Sequence C - Whole Face (6 min.)	
└ Neutral → FEAR → DISGUST → SURPRISED → Neutral	
(2× repeated)	
REST (1 Minute)	
BLOCK 4: Sequence D - Special Exercises (3 min.)	
└ Neutral → PUFF CHEEKS → SMILE+CHEEKS	
→ Neutral (2× repeated)	

COOL DOWN (2 Minutes)	
-----------------------	--

SESSION COMPLETION (1 Minute)	
"You mastered 4 complex sequences today!"	
"Success rate: 88%"	
"Transition quality: Excellent"	
"25-second hold duration achieved - outstanding!"	

TOTAL DURATION: 25-27 Minutes

Mastery Criterion Phase 3: - ✓ ≥85% of sequences correct - ✓ Average hold duration ≥20 seconds (80% of 25 sec.) - ✓ Smooth transitions → **PROGRESSION to Phase 4**

If Criterion Not Met: - Repeat Week 4 - Optional: 3-emotion sequences instead of 4-5 - Optional: Return to Phase 2 if <70% success

Phase 4: Mastery (Week 5-8)

Goal: Automatization, targeted optimization of weak points

Parameters: - **Frequency:** 3× per week - **Hold duration:** 25 seconds - **Session duration:** 25-27 minutes - **Exercise type:** Flexible mix (60% sequences + 40% isolated exercises)

Structure:

60% SEQUENCES (15 Minutes)	
• 2 complex sequences from Phase 3 • Each repeated 2× times • Focus: Quality and automatization	

40% ISOLATED EXERCISES (8-10 Minutes)	
• System identifies 3-4 problematic exercises • Example: "Disgust" and "Sadness" <85% success • Targeted Blocked Practice: - DISGUST: 25 sec., 4× (3 min.) - SADNESS: 25 sec., 4× (3 min.) - FEAR: 25 sec., 3× (2 min.)	

Adaptive Logic:

IF all exercises ≥90% success:

- 100% sequences (no targeted practice)
- Introduction of new, more challenging sequences

ELSE:

- 60% sequences + 40% targeted practice on weak points

Mastery Criterion Phase 4: - ✓ ≥90% success rate over 4 weeks - ✓ All 9 expressions mastered - ✓ Sequences executed fluently → **TRANSITION to Maintenance Phase**

Maintenance Phase (From Week 9)

Goal: Long-term retention without burnout

Parameters: - **Frequency:** 2-3× per week - **Hold duration:** 25 seconds - **Session duration:** 15-20 minutes (shortened) - **Exercise type:** Flexible mix

Structure:

SHORT SESSION (15-20 Minutes)	
• Warm-up: 2 min. • 1-2 complex sequences: 10-12 min. • 2-3 isolated exercises: 4-6 min.	

- Cool down: 2 min.

Flexibility: - **Short days** (15 min.): 1 sequence + 2 individual exercises - **Intensive days** (25 min.): 3 sequences + targeted individual exercises

Long-term Monitoring: - **Monthly self-evaluation:** Perform all 9 exercises - **If <80% success:** 2-week refresher phase (daily 20 min.) - **Quarterly:** Optional therapist consultation

Part 3: Progression Logic and Decision Algorithm

Definition: "Correct Execution"

An exercise is considered correct when:

1. **Hold duration:** $\geq 80\%$ of target hold duration achieved
 - 15 sec. target $\rightarrow \geq 12$ sec. acceptable
 - 20 sec. target $\rightarrow \geq 16$ sec. acceptable
 - 25 sec. target $\rightarrow \geq 20$ sec. acceptable
2. **Muscle activation:** Target musculature visibly activated
 - Avatar evaluates visually (optional: camera-based)
3. **Form accuracy:** Expression matches model to $\geq 70\%$
4. **No compensation:** No excessive compensatory movements

Weekly Evaluation Logic

FUNCTION evaluate_weekly_progress(week, success_rate, avg_hold_duration):

```
IF week <= 2: # Phase 1
    target_hold = 15
    target_success = 0.80
    next_phase = "Phase 2: Antagonistic Pairs"

ELIF week == 3: # Phase 2
    target_hold = 20
    target_success = 0.85
    next_phase = "Phase 3: Complex Sequences"

ELIF week == 4: # Phase 3
    target_hold = 25
    target_success = 0.85
    next_phase = "Phase 4: Mastery"

ELSE: # Phase 4+
```

```

    target_hold = 25
    target_success = 0.90
    next_phase = "Maintenance (2-3x/week)"

# Checks
hold_achieved = (avg_hold_duration >= target_hold * 0.8)
success_achieved = (success_rate >= target_success)

# Decision
IF hold_achieved AND success_achieved:
    RETURN "✓ PROGRESSION to " + next_phase

ELIF success_rate >= (target_success - 0.10): # Borderline
    RETURN "△ MODIFIED REPETITION"

ELSE: # <70% success
    RETURN "☹ FULL REPETITION"

```

Progression Scenarios

Success Rate	Hold Duration	Decision	Action
≥90%	≥90%	✓ Excellent	Progression + optional accelerated advancement
80-89%	80-89%	✓ Mastery	Progression to next phase
70-79%	70-79%	△ Borderline	Modified repetition (reduced parameters)
60-69%	60-79%	☹ Not Achieved	Full repetition with enhanced support
<60%	<70%	△ Critical	Break + therapist consultation recommended

Part 4: Avatar System Implementation

Required Animation Sequences (111 Total)

Base Expressions (18 Sequences)

- 9 exercises × 2 variants (15 sec. + 25 sec. version) = 18

Demonstrations (18 Sequences)

- 9 exercises × 2 speeds (slow + normal) = 18

Pair Transitions (8 Sequences)

- Surprised ↔ Angry (2)
- Smile ↔ Sadness (2)
- Disgust ↔ Neutral (2)
- Fear ↔ Surprised (2)

Sequence Transitions (37 Sequences)

- Neutral → each emotion (9)
- Each emotion → Neutral (9)
- Sequence demonstrations (4)
- Frequent intermediate transitions (15)

Feedback & Support (30 Sequences)

- Positive feedback (5 variants)
- Correction feedback (5 variants)
- Motivational feedback (5 variants)
- Transition sequences (10)
- Special sequences (5)

Real-time Feedback During Hold Phase

0-5 Seconds (Build-up Phase): - “Form the expression completely” - Progress bar starts

5-20 Seconds (Main Hold): - “Very good, maintain the position” - “10 seconds remaining” - Visual: Green frame = correct

20-25 Seconds (Finalization): - Countdown: “5, 4, 3, 2, 1” - “Perfect! 25 seconds held!”

On Early Release: - “You held for 18 seconds - great start!” - “Next time we’ll try for 20 seconds” - No negative feedback

Adaptive Support

Level 1: Minimal Support (≥85% success) - Avatar demonstrates once only - Brief verbal cues - Patient works autonomously

Level 2: Moderate Support (70-84% success) - Avatar demonstrates before each exercise - Real-time visual feedback - Encouraging cues

Level 3: Intensive Support (<70% success) - Step-by-step guidance - Avatar demonstrates simultaneously (mirroring) - Focus on fewer exercises - Therapist support recommended

Part 5: Integration into Existing Therapy

Realistic Contextualization

Parkinson’s patients typically already engage in: - Physical therapy: 1-2×/week, 45-60 min. - Sport/Walking: 2-3×/week, 30-45 min. - Potentially: LSVT, Speech therapy, other therapies

Facial exercises = SUPPLEMENT, not replacement

Optimal Training Days

EXAMPLE WEEKLY PLAN:

Monday: Physical Therapy (60 min.)
 → NO facial training

Tuesday: Facial Training (20 min.) ✓

Wednesday: Sport/Walking (45 min.)

Thursday: Facial Training (20 min.) ✓

Friday: Physical Therapy (60 min.)
 → NO facial training

Saturday: Facial Training (20 min.) ✓

Sunday: Rest day or optional training

→ Total: 3× facial training, integrated into routine

Onboarding Questions

Avatar asks at program start:

“What other therapies/exercises are you currently doing?” - ☐ Physical therapy (**× per week**) - ☐ **Speech therapy/LSVT LOUD** - ☐ **Sport/Fitness** (× per week) - ☐ Other:

System Recommendation Based on Total Load:

IF existing therapy <120 min./week:

→ Recommendation: 4-5× facial training/week

ELIF 120-240 min./week:

→ Recommendation: 3-4× facial training/week

ELSE (≥240 min./week):

→ Recommendation: 2-3× facial training/week

→ "You're already very active! Facial exercises supplement your program."

Part 6: Adherence Promotion

Critical Success Factors

1. **Automatic Reminders** - Daily push notification at chosen time - “Forgetting” is primary barrier - reminders essential
2. **Streak Tracking** - “You’ve trained 7 days in a row! 🔥” - Visualization of consistency
3. **Weekly Summary** - Email with progress to patient (optional: to family) - Graphical weekly performance display
4. **Quick Session for Bad Days** - 10-min. version with 3-4 core exercises - Avatar: “Today 10 minutes is better than 0!” - Counts as “completed” for streak
5. **Flexible Rest Days** - Patient chooses 1-2 rest days/week themselves - Avatar: “Enjoy your rest day!”
6. **Personalization Based on Daily Form**

Check before session: "How are you feeling today?"
[Very good / Good / OK / Not so good]

IF "Not so good":

- Reduce hold duration (-5 sec.)
- Longer rests
- Offer quick session

Expected Adherence Rates

Based on evidence and realistic frequency (3-4×/week):

- **Weeks 1-4:** 85-90% adherence (honeymoon period)
- **Weeks 5-8:** 75-85% adherence (realistic adjustment)
- **Maintenance:** 70-80% adherence (long-term)

→ **Significantly higher than daily programs (47%)**

Part 7: Therapist Dashboard

Monitoring Metrics

Daily Captured: - Success rate per exercise (9 data points) - Average hold duration - Number of attempts until correct execution - Session timing (on/off period)

Weekly Display: - Overall success rate (%) - Progression vs. regression - Adherence rate (completed sessions / planned sessions) - Problem exercises (consistently <80%)

Dashboard View for Therapists:

PATIENT: Max Mustermann CURRENT PHASE: Phase 2 (Week 3) SUCCESS RATE: 82% ADHERENCE: 11 of 12 sessions (92%)	
PROBLEM EXERCISES: • Disgust: 65% (needs attention) • Sadness: 72% (below target)	
RECOMMENDATION: Repeat Week 3 with focus on "Disgust" and "Sadness"	

Alarm System: - Success rate <70%: Red - Adherence: >2 days without activity: Yellow
- No training for 1 week: Red → Therapist contacts patient

Part 8: Evidence Base and References

Motor Learning

- Blocked vs. Random Practice: [web:23][web:97][web:100][web:103]
- Contextual interference: [web:108][web:111]
- Progressive practice: [web:34]

Parkinson-Specific

- Facial exercises: [web:3][web:6][web:9]
- LSVT protocols: [web:165][web:167][web:170]
- Home exercise adherence: [web:177][web:179][web:182]

Hold Duration and Dosing

- Flexibility standards: [web:137]
- Facial muscle training: [web:145][web:153][web:159]
- Parkinson exercise guidelines: [web:19][web:25][web:31]

Digital Interventions

- Technology-supported programs: [web:70][web:180][web:192]
 - Adherence with digital tools: [web:64][web:67][web:196]
-

Part 9: Quick-Reference Tables

Overview: All 5 Phases

Parameter	Phase 1	Phase 2	Phase 3	Phase 4	Maintenance
Weeks	1-2	3	4	5-8	9+
Frequency	3-4×/wk	3-4×/wk	3-4×/wk	3×/wk	2-3×/wk
Hold Duration	15 sec.	20 sec.	25 sec.	25 sec.	25 sec.
Exercise Type	Isolated	Pairs	Sequences	Mix	Flexible
Session	18-20 min.	22-24 min.	25-27 min.	25-27 min.	15-20 min.
Weekly Load	54-80 min.	66-96 min.	75-108 min.	75 min.	30-60 min.
Mastery	≥80%	≥85%	≥85%	≥90%	≥85%

Antagonistic Pairs (Phase 2)

Pair	Emotion 1	Emotion 2	Antagonism
1	Surprised	Angry	Raise forehead ↔ furrow forehead
2	Smile	Sadness	Mouth corners up ↔ down
3	Disgust	Neutral	Activation ↔ relaxation
4	Fear	Surprised	Eyes squeezed ↔ wide

Complex Sequences (Phase 3)

Sequence	Emotions	Duration	Region
A	Neutral → Surprised → Angry → Surprised → Neutral	2.5 min.	Upper face
B	Neutral → Smile → Sadness → Pout → Smile → Neutral	3 min.	Lower face
C	Neutral → Fear → Disgust → Surprised → Neutral	2.5 min.	Whole face
D	Neutral → Puff cheeks →	1.5 min.	Special exercises

Sequence	Emotions	Duration	Region
	Smile+Cheeks → Neutral		

Progression Criteria

From Phase	To Phase	Criteria
1 → 2	Foundations → Consolidation	≥80% success + ≥12 sec. hold
2 → 3	Consolidation → Intensification	≥85% success + ≥16 sec. hold + smooth transitions
3 → 4	Intensification → Mastery	≥85% success + ≥20 sec. hold
4 → Maintenance	Mastery → Maintenance	≥90% success over 4 weeks
